

The oocyte zinc transporter *Slc39a10/Zip10* is a regulator of zinc sparks during fertilization in mice

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eLife Assessment

This study presents significant and novel insights into the roles of zinc in mammalian meiosis/fertilization events. These findings are **useful** to our understanding of these processes. The evidence presented is **solid**, with experiments being well-designed, carefully described, and interpreted with appropriate rigor. The main limitation of lack of mechanistic insight needs to be acknowledged.

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Abstract

In all vertebrates studied to date, a rise(s) in intracellular calcium is indispensable for successful fertilization and further embryonic development. Recent studies demonstrated that zinc is ejected to the extracellular milieu, the ‘zinc spark’, and follows the first few calcium rises of fertilization. However, the role of the zinc sparks in fertilization and development, and the supporting influx mechanism(s) are unknown. In this study, we focused on zinc transporters *Zip10/Slc39a10* which was expressed in mouse oocytes through follicular development, and investigated the oocyte-specific deficient mice for *Zip10* (*Zip10^{d/d}*; *Zip10^{flox/flox} Gdf9^{Cre/+}*). *Zip10* mRNA or ZIP10 protein was expressed throughout folliculogenesis in the oocyte or plasma membrane, respectively. The number of ovulated oocytes was examined in *Zip10^{d/d}* mice, and no change from the number of oocytes was observed. *Zip10^{d/d}* oocytes decreased zinc level in the oocytes, but did not affect maturation

and metaphase II spindles formation. Fertilization-induced calcium oscillations were present in *Zip10^{d/d}* oocytes, but zinc sparks were not observed. Despite other events of egg activation proceeding normally in *Zip10^{d/d}* oocytes, embryo development into 4-cells and beyond was compromised. We show here for the first time that the zinc transporter ZIP10 contributes to zinc homeostasis in oocytes and embryos, highlighting the role of labile zinc ions in early development.

Impact statement

The zinc transporter, *Slc39a10/Zip10*, is required for the zinc sparks of fertilization in mice.

Introduction

In mammalian fertilization, the sperm factor is released into the ooplasm to activate the oocyte (Yoon and Fissore, 2007 [\[1\]](#)). It is now believed that phospholipase C zeta (PLC ζ) (Yoon and Fissore, 2007 [\[1\]](#); Saunders et al., 2002 [\[2\]](#)) is the sperm factor and it induces in the ooplasm repetitive increases in the intracellular Ca²⁺, termed ‘calcium oscillations.’ These oscillations play an essential role in triggering the oocyte activation events, such as cortical granule exocytosis, the block of polyspermy, meiotic resumption and exit from MII arrest (Xu et al., 1994 [\[3\]](#); Jones, 2005 [\[4\]](#); Ducibella et al., 2006 [\[5\]](#); Ito et al., 2011 [\[6\]](#); Sugita et al., 2024 [\[7\]](#)). A calcium increase is highly conserved in many species as the trigger of egg activation and is vital for successful fertilization.

In recent years, zinc ions as well as calcium ions, are thought to play important roles during fertilization (Suzuki et al., 2010 [\[8\]](#); Allouche-Fitoussi and Breitbart, 2020 [\[9\]](#); Kageyama et al., 2022 [\[10\]](#)). Zinc ion is an essential trace element and the second most abundant transition ion in the human body after iron (Lubna and Ahmad, 2023 [\[11\]](#)). Zinc homeostasis is essential for optimal metabolic function in the reproductive process in mammals (Pascua et al., 2020 [\[12\]](#)), therefore zinc deficiency causes abnormalities such as fetal teratogenicity, long gestation periods, problematic labor, low birth weight and weak offspring (Favier, 1992 [\[13\]](#); Bedwal and Bahuguna, 1994 [\[14\]](#)). Zinc also has important role in the male and female germ cells (Allouche-Fitoussi and Breitbart, 2020 [\[9\]](#); Kageyama et al., 2022 [\[10\]](#)). Especially in the mouse female germ cells, zinc-deficient condition caused profound defects in oocyte maturation, cumulus cell expansion and ovulation cycle (Tian and Diaz, 2012 [\[15\]](#)). Several studies have been conducted in mice to understand how zinc regulates oocyte maturation. Zinc was reported to be a critical regulator of meiosis throughout oocyte maturation, including maintaining release from the first and second meiotic arrest (Kong et al., 2012 [\[16\]](#)). In addition, zinc levels of oocytes were found to increase by 50% as the oocyte progressed through meiosis (Kim et al., 2010 [\[17\]](#); Bernhardt et al., 2011 [\[18\]](#)).

In 2011, Kim *et al.* reported the importance of zinc during mammalian fertilization (Kim et al., 2011 [\[19\]](#)). Zinc is predominantly stored in vesicles that are symmetrically arranged along the oocyte cortex at the GV stage, and zinc-stored vesicles (cortical granules) are located away from the spindle and form a hemispherical pattern at the MII stage (Kim et al., 2011 [\[19\]](#); Que et al., 2015 [\[20\]](#); Kong et al., 2014 [\[21\]](#); Que et al., 2019 [\[22\]](#); Jo et al., 2019 [\[23\]](#); Que et al., 2017 [\[24\]](#)). After the penetration of sperm, it has been induced a ‘zinc spark’ that releases billions of labile zinc ions from the cortical granules, and the total zinc content of the oocytes decreases by 10-20% in at the end of fertilization in mice (Kim et al., 2011 [\[19\]](#); Lee et al., 2020 [\[25\]](#)). In addition, these events immediately follow a series of calcium oscillations (Kim et al., 2011 [\[19\]](#)). Zinc spark has also been observed in the cattle, non-human primate, human and *Xenopus laevis* oocytes (Kim et al., 2011 [\[19\]](#); Que et al., 2019 [\[22\]](#); Duncan et al., 2016 [\[26\]](#); Seeler et al., 2021 [\[27\]](#)), suggesting that this phenomenon is a highly conserved event in vertebrate. More surprisingly, full-term development of mouse embryos has been reported by chelation of Zn²⁺ ions without Ca²⁺ release, suggesting the depletion of zinc ions

in oocytes may be sufficient for oocyte activation (Suzuki et al., 2010). Zinc has also been reported to impact the sperm during fertilization in mice. Zinc accumulation in the zona pellucida increases fibril binding along the glycoprotein matrix and decreases the number of sperm that can reach the fertilized oocytes (Que et al., 2017; Aonuma et al., 1981; Tokuhiko and Dean, 2018). After the zinc sparks, the released zinc affects the forward motility of sperm to prevent their passage through the zona matrix (Tokuhiko and Dean, 2018). Zinc spark is associated with the release of ovastacin (*Astl*, the official gene name), which is required for the zona reaction, and functions as a polyspermy block mechanism that is initiated only a few minutes after fertilization (Tokuhiko and Dean, 2018).

The transport of zinc ions into and out of cells is regulated by zinc transporters. In mammals, 14 zinc ion importers, called as ZIP (ZIP1-ZIP14), have been identified (Fukada and Kambe, 2011; Takagishi et al., 2017; Kambe et al., 2015). Previous study showed that mammalian oocytes regulate zinc uptake through two maternally derived and cortically distributed zinc transporters, ZIP6 and ZIP10 (Kong et al., 2014). Further, they reported that targeted disruption using *Zip6*- and *Zip10*-specific morpholino injection or antibody incubation induced alteration of the intracellular labile zinc content, spontaneous resumption of meiosis from the PI arrest and premature arrest at a telophase I-like state (Kong et al., 2014). It is clear from these reports that ZIP6 and ZIP10 are involved in zinc transport in oocytes, but the function is not elucidated.

In this study, we generated oocyte specific *Zip6* and *Zip10* conditional knockout mice and examined the function of ZIP10 in the oocytes and the importance of zinc homeostasis during fertilization and embryonic development. This study provides clues that elucidate its role in fertilization and embryonic development, which is still largely unknown. In addition, this is the first report confirming the function of the zinc transporter in oocytes, which will contribute to future research.

Results

Zinc transporters, ZIP6, and ZIP10, are expressed in mouse oocytes through follicular development

First, using *in situ* hybridization and immunofluorescent staining, we examined the expression of ZIP10 during follicular development. As shown in Fig. 1A, *Zip10* mRNA was expressed from the primordial oocyte (Arrow). It continued to be expressed in oocytes of primary, secondary, and antral follicles. ZIP10 protein was also expressed in the plasma membrane of primordial oocytes (Fig. 1B; Arrow). We also confirmed the expression of ZIP10 protein at the plasma membrane of oocytes of primary, secondary, and antral follicles. Although ZIP6 was also expressed in oocytes throughout folliculogenesis (Fig. 1C; Arrow), it displayed nuclear localization in oocytes and granulosa cells of primary, secondary, and antral follicles. Fig. 1D showed zona pellucida and granulosa cells through the follicular development.

Phenotype of oocyte-specific *Zip6*^{d/d} and *Zip10*^{d/d} female mice

To elucidate the roles of *Zip6* and *Zip10* in the mouse oocytes, oocyte-specific *Zip6* (*Zip6*^{d/d}) and *Zip10* (*Zip10*^{d/d}) knockout mice were generated (Fig. S1A-C). We examined whether ZIP6 and ZIP10 protein expression was absent in *Zip6*^{d/d} and *Zip10*^{d/d} oocytes (Fig. S1D). *Zip6*^{ff} and *Zip10*^{ff} mice were used as controls, respectively. We examined the number of ovulated oocytes or defects in oocyte maturation. After superovulation, the number of ovulated oocytes collected from the oviduct in *Zip10*^{d/d} mice was 22.1 ± 2.5 oocytes/mouse (Fig. 2A), which was equivalent to that in *Zip10*^{ff} mice (18.7 ± 2.9 oocytes/mouse, *p* > 0.05). The numbers of ovulated oocytes in *Zip6*^{d/d} and *Zip6*^{ff} mice were also equivalent (28.7 ± 3.6 and 32.8 ± 3.3 oocytes/mice, respectively, *p* > 0.05) (Fig.

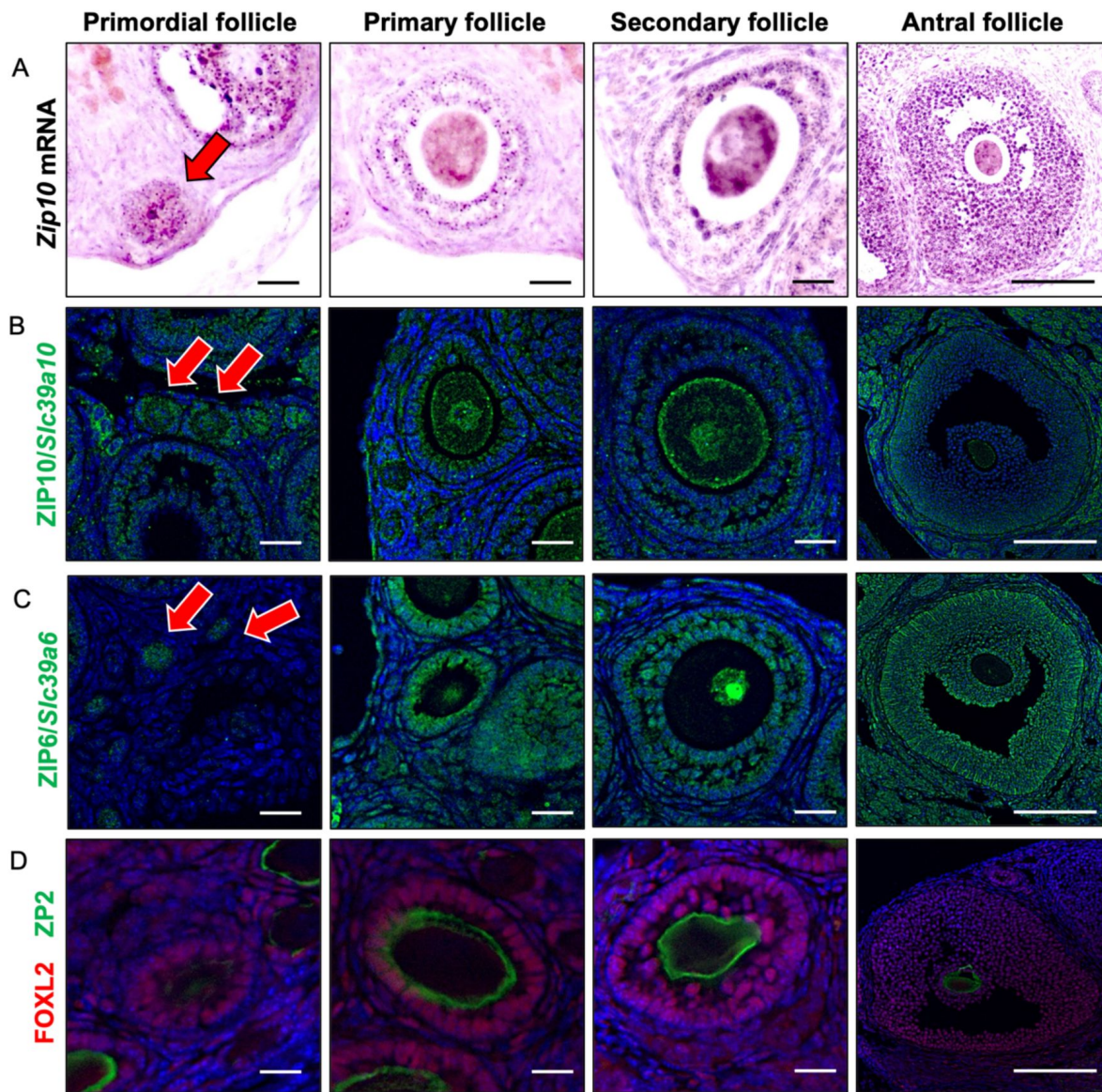


Fig. 1.

Expression of ZIP6 and ZIP10 in mouse ovary.

(A) In situ hybridization in the mouse ovary showed ZIP10 expression in oocyte and granulosa cell from primordial, primary, secondary and antral follicle. Arrow indicates primordial follicular oocyte. (B) Immunofluorescent staining for ZIP10 (green) in the mouse ovary showed ZIP10 expression in oocyte membrane. Arrow indicates primordial follicular oocyte. (C) Immunofluorescent staining for ZIP6 (green) in the mouse ovary showed ZIP6 expression in oocyte nucleus and granulosa cells. Arrow indicates primordial follicular oocyte. (D) Immunofluorescent staining showed ZIP2 (green; zona pellucida) and FOXL2 (red; granulosa cells) in the mouse ovary. It was observed that ZIP2 was not present in the primordial follicle; however, it was present in the primary, secondary and antral follicles. Furthermore, FOXL2 was observed at granulosa cells of all stage follicles. Scale bar: 20 μ m (primordial, primary and secondary follicle); 150 μ m (antral follicle) (A-D).

S2A). As for oocyte maturation, the rate of oocytes with a first polar body at 10h, 12h, and 14h in the *Zip10^{d/d}* group was not different than that of *Zip10^{ff/ff}* group (*Zip10^{d/d}*: 58.7%, 61.9%, and 79.4%, *Zip10^{ff/ff}*: 50.8%, 54.1% and 70.5%, respectively, $p > 0.05$, **Fig. 2B**). In addition, *Zip10^{d/d}* oocytes display intact metaphase spindles just as in the *Zip10^{ff/ff}* groups (**Fig. 2C**).

Next, we examined the levels of labile zinc in germinal vesicle (GV), MII, and two pronuclei (2PN) zygotes by comparing the fluorescence intensity following loading with the dye, FluoZin-3AM (**Fig. 2D**). The FluoZin-3AM fluorescence intensity in GV and MII oocytes of the *Zip10^{d/d}* group was lower than the *Zip10^{ff/ff}* group ($p < 0.05$). After fertilization, the fluorescence intensity in *Zip10^{ff/ff}* zygotes decreased dramatically, and also decreased for the *Zip10^{d/d}* group, although it remained higher than for *Zip10^{ff/ff}* oocytes. When compared within group, the fluorescence intensity in *Zip10^{ff/ff}* between GV and MII oocytes was significantly different (GV vs. MII and MII vs 2PN; $p < 0.05$, respectively), but the stages of *Zip10^{d/d}* oocytes were not different despite clear trends ($p > 0.05$). The levels of zinc fluorescence intensity in the *Zip6^{d/d}* group were not different from the controls *Zip6^{ff/ff}*, which decreased markedly at fertilization (**Fig. S2B**; $p > 0.05$).

The zinc sparks of *Zip10^{d/d}* oocytes were suppressed after fertilization or artificial activation

To determine if the absence of ZIP6 and ZIP10 influenced the detection of Zn^{2+} sparks associated with fertilization, we monitored extracellular zinc and intracellular calcium during fertilization of *Zip6^{d/d}* and *Zip10^{d/d}* oocytes (Fig. S3 and 3A). *Zip10^{ff/ff}* oocytes displayed the expected calcium oscillations following fertilization, and a zinc spark followed the first calcium rise (**Fig. 3A** upper side, **movies 1 and 2**). In contrast, *Zip10^{d/d}* oocytes did not release zinc ions immediately after the first calcium spike, despite mounting normal calcium oscillations (**Fig. 3A** lower side, **movies 3 and 4**). In *Zip6^{d/d}* oocytes, a zinc spark occurred immediately after the first intracellular calcium rise at fertilization, just as in control *Zip6^{ff/ff}* oocytes (Fig. S3, **movie S1-S4**). The extracellular zinc sparks were examined following artificial oocyte activation of mouse oocytes with ionomycin. The zinc sparks occurred immediately after the intracellular calcium rise in *Zip10^{ff/ff}* oocytes (**Fig. 3B** upper side, **movies 5 and 6**), however, *Zip10^{d/d}* oocytes did not release zinc ions after the calcium spike (**Fig. 3B lower side**, **movies 7 and 8**).

Zip10^{d/d} mouse oocytes can be fertilized but were unlikely to develop to the blastocysts

Zip6^{d/d} and *Zip10^{d/d}* oocytes then were used for IVF, and we examined the rates of fertilization, polyspermy, and embryo development. Successful fertilization was confirmed at 6 h after IVF by the presence of pronuclei (**Fig. 4A** and **B**). *Zip10^{d/d}* oocytes were fertilized at rates like those observed for control oocytes (**Fig. 4A**). Consistent with this, the zona reaction triggered by fertilization and examined using a ZP2 antibody was similar in *Zip10^{d/d}* and *Zip10^{ff/ff}* oocytes (**Fig. 4D**). We also examined the localization of ovastacin, whose expression in the cortex and loss post-fertilization was comparable between the two groups (**Fig. 4E**). As for the expression of JUNO, it had the same expression than between null and control oocytes (Fig. S4) and the temporal dynamics of its disappearance from the cortex after fertilization was similar for both *Zip10^{ff/ff}* and *Zip10^{d/d}* groups (**Fig. 4F**).

We next examined the development to the blastocyst stage, and whereas *Zip10^{ff/ff}* zygotes developed at the expected rates, approximately 78.9% (2 cell), 90.3 (4-8 cell) and 75.0% (blastocyst), a smaller fraction of *Zip10^{d/d}* zygotes did 37.2% (2 cell) and 38.3% (4-8 cell) and only 32.7% reached blastocyst stage ($p < 0.05$; **Fig. 4B**). As shown in **Fig. 4C**, the total cell numbers of blastocysts were also lower for *Zip10^{d/d}* embryos (51.6 ± 2.1 cells) than for those of *Zip10^{ff/ff}* ones (72.7 ± 2.1 cells, $p < 0.05$). *Zip6^{d/d}* oocytes did not display any alterations in the rate of fertilization or development (**Fig. S2C and D**; $p > 0.05$).

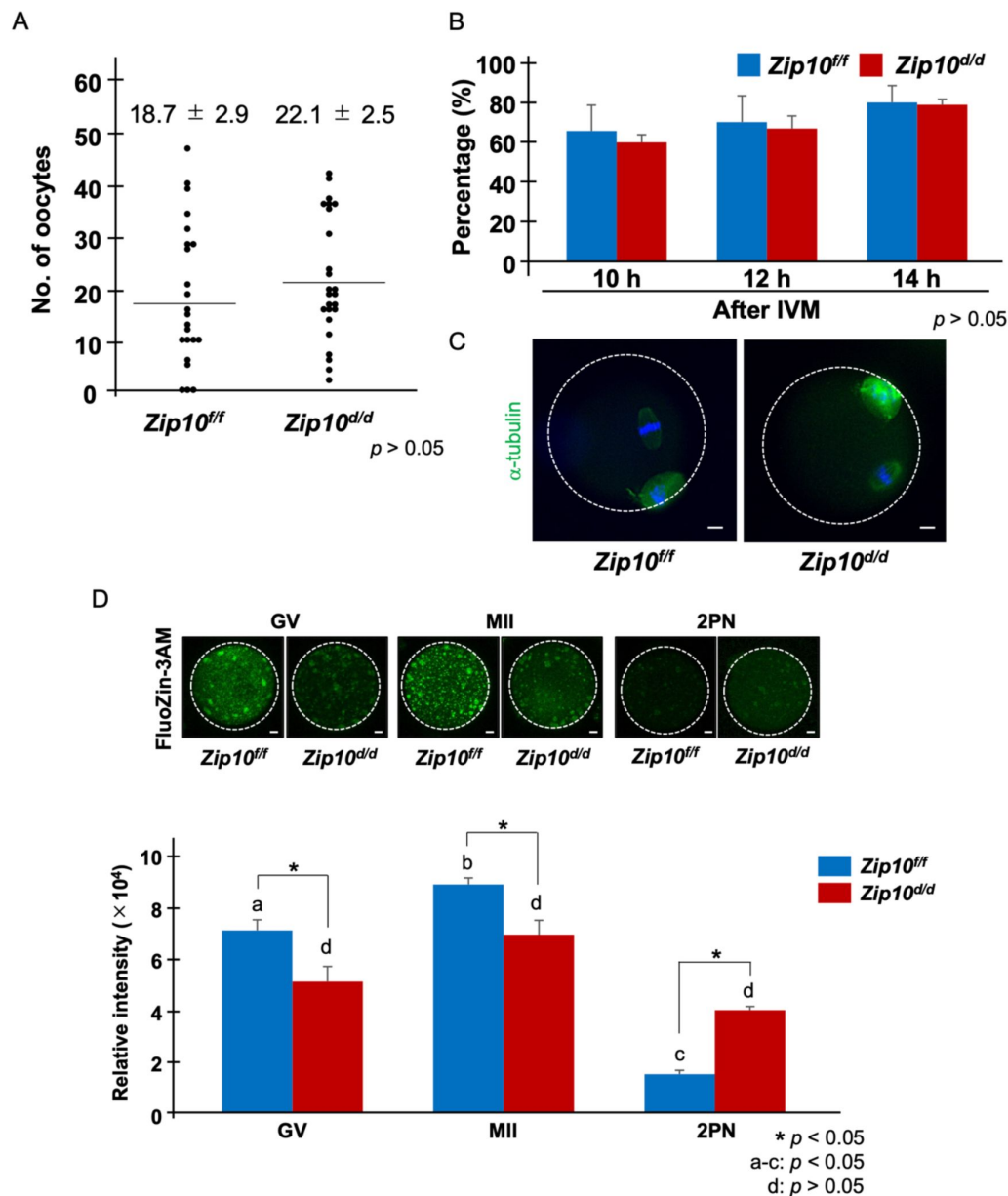


Fig. 2

Number of collected oocytes and dynamics of labile zinc ion in *Zip10^{d/d}* mice.

(A) The results of average number of oocytes in each group. Data represents the average $\pm$ SEM. These experiments were repeated at least thrice. Statistical differences were calculated according to student's *t*-test ($p > 0.05$; no significant difference). (B) The percentage of extrusion of first polar body at 10, 12 and 14 h after IVM. These experiments were repeated at least thrice. Statistical differences were calculated according to student's *t*-test ($p > 0.05$; no significant difference). (C) The morphology of spindle and chromosome organization in *Zip10^{ff}* and *Zip10^{d/d}* MII oocytes at 14 h after IVM. Anti- α -tubulin antibody (green) was used to stain the spindles. Chromosomes were stained with DAPI (blue). The scale bar represents 10 μ m. (D) Comparison with the fluorescence intensity of intracellular labile zinc ion in GV, MII and 2PN. The upper images showed the fluorescence of intracellular labile zinc ion of oocyte or embryo treated with 2 μ M FluoZin-3AM for 1h. Representative images are shown. The white dotted circles indicate the positions of oocytes and embryos. Scale bars denote 10 μ m. The lower part showed the fluorescence intensity of labile zinc ions in oocytes or embryos. Data represent the average $\pm$ SE of the experiments. For each experiment, 10–20 oocytes/embryos were stained and used for the measurement in each stage of the experiment, and these experiments were repeated three times. Statistical differences were calculated according to the welch's *t*-test. Different letters represent significant differences ($p < 0.05$).

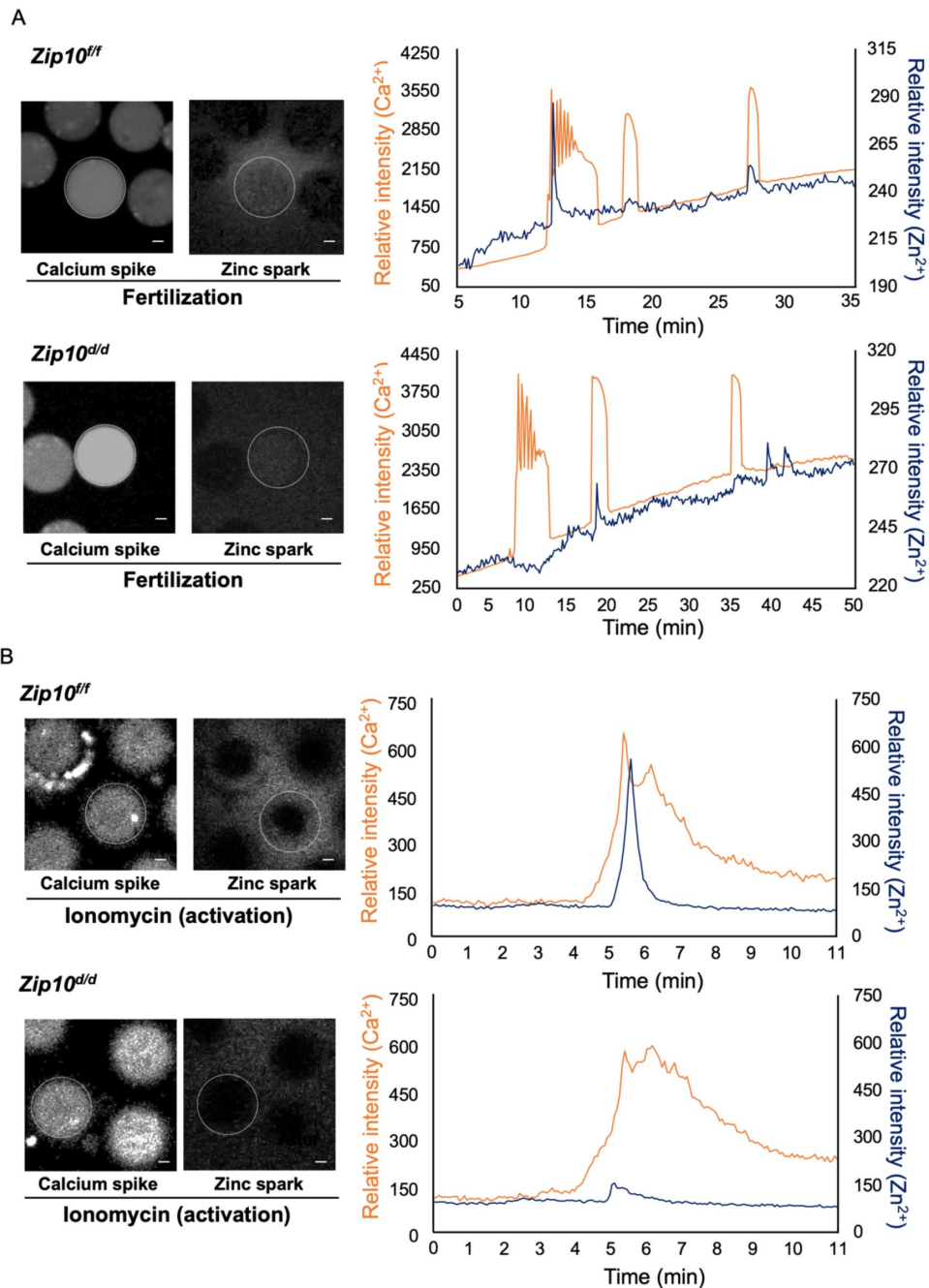


Fig. 3

Measurement of calcium spike and zinc spark in *Zip10^{d/d}* mice.

(A) The representative images of calcium spike and zinc spark after IVF in mouse oocytes. Left side images showed calcium spike. Right side images showed zinc spark. The oocytes increased calcium ion and released zinc ion shortly after fertilization. The white dotted circles indicate the positions of oocytes. Successful fertilization was confirmed by simultaneously monitoring intracellular calcium oscillations with Calbryte 590 AM and extracellular zinc ions with FluoZin-3 every 4 s. Capacitated frozen-thawed sperm was added to MII at 2 min after imaging start. Orange line showed calcium ion and dark blue line showed zinc ion. Intracellular calcium increases immediately before a zinc spark. Scale bars denote 20 μ m. (B) The representative images of an MII egg activated with 5 μ M ionomycin followed by monitoring of intracellular calcium oscillations with Calbryte 590 AM and extracellular zinc using 20 μ M FluoZin-3. The ionomycin was added to MII at 2 min after imaging start. Orange line showed calcium ion and dark blue line showed zinc ion. Intracellular calcium increases immediately before a zinc spark. Scale bars denote 20 μ m.

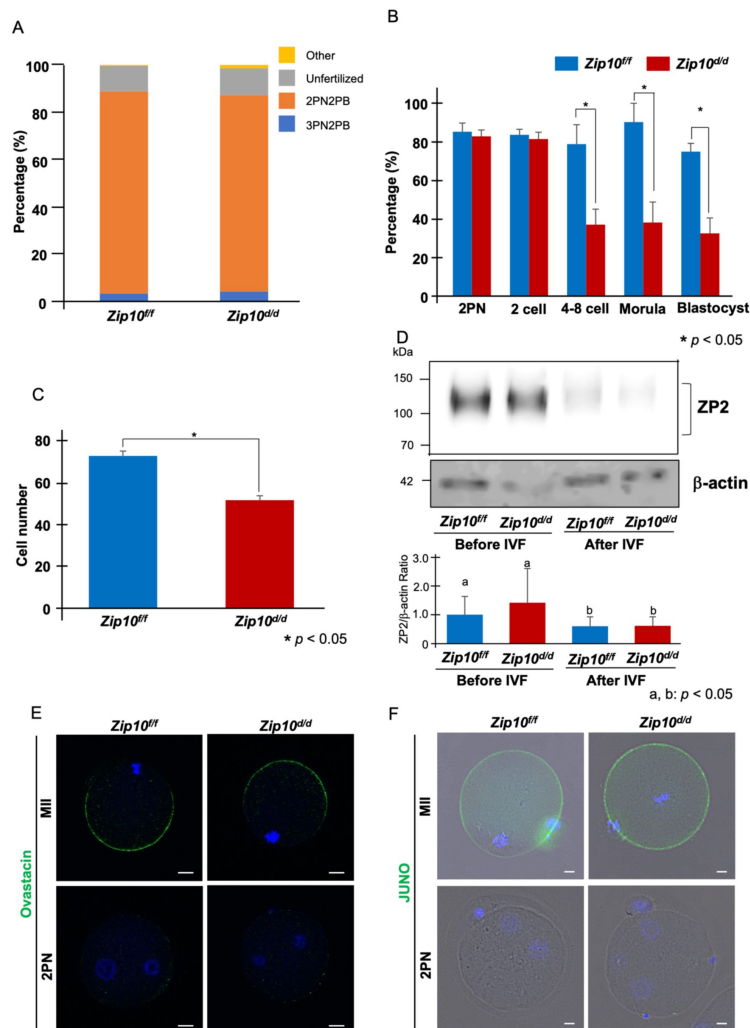


Fig. 4

Presence of a mechanism to prevent polysperm fertilization in *Zip10* cKO mice.

(A) The percentages of oocytes with each number of PN at 6 h after insemination. Yellow region showed other including degeneration, degression and fragmentation. Gray region showed unfertilization, namely MII oocytes. Orange showed 2PN2PB, namely embryo possessed one female and male pronucleous (2PN) and second polar body (2PB). Blue region showed polysperm fertilization (3PN2PB). (B) The percentage of fertilized oocytes and developmental embryos. Data represent the average $\pm$ SE of the experiments. The embryo development was observed at 6 (2PN) 24 (2 cell), 48 (4-8 cell), 72 (Morula), and 96 (Blastocyst) hours after IVF. The oocytes used for IVF were calculated as the parameter for the fertilization rate and the rate of embryo development. These experiments were repeated at least thrice. Statistical differences were calculated according to the chi-square test. Different letters represent significant differences ($p < 0.05$). (C) The cell number of blastocyst derived from IVF. Blastocysts were fixed, immunostained, and physically flattened between a slide and coverslip. Photographs represent a single plane of focus. Nuclei are indicated by DAPI staining. The count used inverted fluorescence microscope. These experiments were repeated three times, and each group counted total 46 embryos. Statistical differences were calculated according to the student's *t*-test ($p < 0.05$; significant difference). (D) Western blot of oocytes from *Zip10^{ff}* and *Zip10^{dd}* mice at 0 or 6 h after insemination using rat anti-ZP2 antibody. Intact ZP2 and the cleaved C-terminal fragment of ZP2 measured 120-130 kDa and undetected, respectively. Expression level of β -actin serves as a protein loading control and quantified the expression level of ZP2. Molecular mass is indicated at the left. Statistical differences were calculated according to the one-way ANOVA. Different letters represent significant differences ($p < 0.05$). (E) MII oocytes and 2PN embryos from *Zip10^{ff}* and *Zip10^{dd}* mice were imaged by confocal microscopy after staining with rabbit anti-ovastacin (green). Chromosomes were stained with DAPI (blue). The scale bar represents 10 μ m. (F) MII oocytes and 2PN embryos from *Zip10^{ff}* and *Zip10^{dd}* mice were imaged by BZ-X700 microscopy after staining with rat anti-mouse FR4 (JUNO; green). Chromosomes were stained with DAPI (blue). The scale bar represents 10 μ m.

Discussion

The zinc sparks associated with fertilization were first discovered in the mouse (Kim et al., 2011). Similar observations followed in other mammalian species and amphibians (Que et al., 2019; Duncan et al., 2016; Seeler et al., 2021). However, the underlying mechanism(s) and biological role of zinc sparks are not elucidated. In this study, we have examined the role of the zinc transporters ZIP6 and ZIP10 using conditionally gene deficient mice, and queries their contribution to the zinc sparks of fertilization.

First, we confirmed the expression of zinc transporter in mouse oocytes. Our results are consistent with previous studies (Kong et al., 2014; Chen et al., 2023). ZIPs, including ZIP6 and ZIP10, are transmembrane proteins (Eide, 2004; Kambe et al., 2004; Lu and Fu, 2007; Schmitt-Ulms et al., 2009). In primordial follicles, the ooplasm staining of ZIP10 we anticipate corresponds to ooplasmic vesicular sites. ZIP10 expression shifted to the plasma membrane in primary and antral follicle phase. A similar localization shift of ZIP10 to the oocyte surface was reported during oocyte maturation, which is much later than reported here by. Furthermore, in that study, ZIP10 was detected in the nuclear/nucleolar positions of oocytes of all follicular stages (Chen et al., 2023), which we did not observe. On the other hand, ZIP6 was expressed at the nuclear/nucleolar regions and in granulosa cells. This localization of ZIP6 was consistent with that in a previous study (Chen et al., 2023). However, our results failed to notice that ZIP6 shifts to the at plasma membrane in antral follicle oocytes (Kong et al., 2014; Chen et al., 2023). The results indicate that ZIP6 may be fulfilling a distinct function within the oocytes compared to ZIP10.

To assess the role of these transporters in mice, we generated oocyte-specific *Zip6* or *Zip10* knockout mice (Fig. S1; *Zip6^{d/d}* and *Zip10^{d/d}*, respectively). Previous reports noted that the disruption of these transporters using specific morpholinos or incubation in function-blocking antibodies induced the change of intracellular labile zinc quota into mice oocytes (Kong et al., 2014). The amount of labile zinc ions in those mouse oocytes was measured using FluoZin-3AM, zinc indicator. The amount of labile zinc ions in *Zip10^{d/d}* oocytes was significantly lower than in the *Zip10^{ff}* (Fig. 2D). We failed to observe any effect on zinc levels of in *Zip6^{d/d}* oocytes (Fig. S2B). The results indicate that ZIP10 is mostly responsible for the uptake of zinc ions in mouse oocytes.

Zinc insufficient GV oocytes do not maintain the meiotic arrest at PI (Kong et al., 2012). This is suggested to happen by premature activation of the MOS-MAPK pathway in the presence of low zinc (Kong et al., 2012). In addition, low ooplasmic zinc accelerates meiotic progression that contributes to the extrusion of large polar bodies (Bernhardt et al., 2011). The early mitotic inhibitor 2 (EMI2), a zinc-binding APC/C proteasome inhibitor, is also an essential component of the cytoplasmic factor (CSF) that initiates entry into MII phase. EMI2 is a zinc binding protein, and when zinc is reduced, the activity of the APC/C proteasome is stimulated, increasing the degradation of CCNB1, reduced MPF activity leading to early release of meiosis arrest (Bernhardt et al., 2011; Bernhardt et al., 2012; Suzuki et al., 2010; Shoji et al., 2014; Ohe et al., 2010). Previous studies reported that targeted disruption of *Zip10* using morpholino injections and function-inhibiting antibodies during meiotic maturation perturbed meiosis progression and resulted in cell cycle arrest at the telophase I-like state (Kong et al., 2014). Based on these findings we hypothesized that the reduced state of labile zinc ions in *Zip10^{d/d}* mouse oocytes may resemble the state of zinc-deficient oocytes. We investigated the maturation progression and spindle organization in oocyte matured in vitro from the GV to MII stage. We failed to observe any differences (Fig. 2A, B and C). We also did not observed abnormalities in oocytes from *Zip6^{ff}* mice (Fig. S2A; $p > 0.05$). We speculate that the disparate outcomes observed in previous studies may be attributed to the presence of trace amounts of labile zinc ions in media or in the oocytes.

Surprisingly, the amount of labile zinc ions in *Zip10^{d/d}*2PN zygote was higher than in *Zip10^{ff}* and despite some differences in fluorescence intensity between *Zip10^{d/d}* GV, MII and 2PN zygote, the labile zinc concentrations were not significantly different between these groups (**Fig. 2D**). Therefore, we monitored zinc sparks and calcium oscillations in the *Zip10*-KO oocytes. Several studies reported the important of IP_3 Rs in mammalian oocytes (Fissore et al., 1999; Parrington et al., 1998), which is essential for egg activation because its inhibition precludes Ca^{2+} oscillations (Miyazaki and Ito, 2006; Miyazaki et al., 1992; Xu et al., 2003). It has been reported that a putative Zinc-finger motif in a helical linker (LNK) domain near the C-terminus of *IP₃R1* plays a role in *IP₃R1* function (Fan et al., 2015; Paknejad and Hite, 2018). Recently, Akizawa et al., reported that both deficient and excessive zinc ions compromise *IP₃R1* sensitivity, diminishing and terminating calcium oscillations (Akizawa et al., 2023). Our results showed that a single zinc spark occurs immediately after first calcium rise of oscillations in *Zip10^{ff}* oocytes, as reported by Kim et al (Kim et al., 2011) (**Fig. 3A**), and similar results are obtained in *Zip6^{d/d}* and *Zip6^{ff}* mouse oocytes (Fig. S3). However, in *Zip10^{d/d}* oocytes, despite the presence calcium elevations, zinc sparks failed to occur. Similarly, *Zip10^{d/d}* oocytes activated with ionomycin (artificial oocyte activation) did not show a zinc spark (**Fig. 3B**). In this study, calcium oscillations occurred in oocytes with low zinc state. In addition, there was no difference in the amplitude frequency of calcium ions in oocytes of both groups within the observation time. Akizawa et al., produced zinc deficiency state with TPEN, a strong chelator of intracellular labile zinc ions (Akizawa et al., 2023). On the other hand, we did not completely remove intracellular labile zinc ions from mouse oocytes. We speculate that zinc ions existed in *Zip10^{d/d}* mouse oocytes induce Ca^{2+} release without compromising *IP₃R1* sensitivity. However, this study demonstrated that the accumulation of intracellular zinc ions mediated by ZIP10 is essential for inducing zinc sparks in mouse oocytes.

The total cellular zinc ions content required for meiosis in the oocyte substantially increases (a 50% increase) from the prophase I arrest to the arrest at metaphase of meiosis II (Kim et al., 2010; Bernhardt et al., 2011). The zinc is stored in undefined cortical granules, and an average of 10^6 zinc atoms are released from these vesicles at the time of fertilization (exocytosis) (Kim et al., 2011; Que et al., 2015; Kong et al., 2014; Que et al., 2019; Jo et al., 2019; Lee et al., 2020). There are several possibilities that might explain why *Zip10* conditional KO mice have low zinc sparks. The most logical is that these oocytes experience reduced zinc loading of cortical of zinc vesicles, leading to the absence of sparks. Our results suggest that in oocytes, zinc sparks and exocytosis require adequate concentrations of labile zinc ions in the cortical secretory vesicles, which is not accomplished when ZIP10 is not present.

Zinc sparks have been reported to contribute to the rapid and permanent block mechanisms to prevent polyspermy. The former is achieved by zinc in the extracellular milieu decelerating the forward motility of sperm (Tokuhiro and Dean, 2018). Secondly, the release of labile pools of zinc at the same time that the zinc metalloendopeptidase ovastacin is released with the cortical granules (Tokuhiro and Dean, 2018; Burkart et al., 2012; Schmitz et al., 2021), cause ZP2 protein cleavage inducing a complete, although belated block to polyspermy (Que et al., 2017; Aonuma et al., 1981; Tokuhiro and Dean, 2018). However, the physiological significance of the zinc sparks' contributions to polyspermy have not been thoroughly tested, and *Zip10^{d/d}* mice, which are almost devoid of them, offer a great model. Our results showing normal rates of monospermic fertilization in *Zip10^{d/d}* oocyte suggest that contributions of Zinc to polyspermy control is negligible (**Fig. 4A**). Our results also show that the ovastacin undergoes normal release in *Zip10^{ff}* oocytes after fertilization (**Fig. 4E** and **4D**), confirming previous studies of a minor role of the ZP2 to polyspermy. We observed the disappearance of JUNO after fertilization (Bianchi et al., 2014) *Zip10^{d/d}*, suggesting that gamete fusion took place (**Fig. 4F**). Together, these results support the view that zinc spark is not directly involved in the polyspermy rejection mechanism.

Interestingly, there were no differences in the rates of fertilization (pronucleus formation), but development to the blastocyst stage was significantly reduced in *Zip10*-null embryos (Fig. 4B). Most of the embryos were arrested at the 2-cell stage and not developed beyond the 4-cell stage. Zinc insufficiency caused altered chromatin structure in the nuclei of blastomeres that displayed decreased global transcription, causing arrest in embryonic development (Kong et al., 2015). The abnormal zinc homeostasis, particularly during the one-cell stage, inhibits the activation of the embryonic genome that occurs around the two-cell stage in mice due to reduce the translational capacity through the inhibition of ribosomal RNA synthesis by RNA polymerase I (Chanfreau, 2013; Garner et al., 2021). Our results also show that *Zip10*^{d/d} blastocysts had significantly lower cell numbers compared to the blastocysts from *Zip10*^{f/f} mice (Fig. 4C). Our results suggest that *Zip10*^{d/d} mice oocytes can be fertilizable, but their developmental potential is decreased. The lack of ZIP10-mediated zinc influx during the folliculogenesis of *Zip10*^{d/d} may compromise gene expression during these early stages of oocyte development. *Zip10*^{d/d} mouse oocytes thus may have compromised developmental potential from the outset. Future studies should assess the transcriptomic profile of *Zip10*^{d/d} mouse oocytes.

In conclusion, we elucidated that ZIP10 is required for zinc ion uptake into oocytes and the intracellular zinc ions regulated by ZIP10 was important for zinc sparks and progression of embryonic development. However, the role of ZIP6 remained uncertain. Currently, the many infertility patients exist in the world (the survey of WHO). It is reported that approximately half of the adult females are deficient in zinc in their serum (Prasad, 1996; Yokokawa et al., 2020). Zinc has been reported to play essential roles in many physiological functions including reproductive functions (Suzuki, M., et al, 2021, MacDonald RS, 2000, Jackson KA, et al, 2008, Bernhardt ML, et al, 2012, Suzuki, T., et al, 2010, Ohe M, et al, 2010). The findings will help elucidate the role of zinc homeostasis in the fields of fertilization/embryogenesis. Furthermore, the development of zinc-focused assisted reproductive technologies and fertilization/embryogenesis media could contribute to improve the developmental potential of oocytes and embryos in other mammalian species.

Materials and methods

All chemicals and reagents were purchased from Sigma-Aldrich (St. Louis, MO, USA) unless otherwise stated.

Animals

Animals were housed in the barrier facility at Azabu University. The deletion of the ZIP10 gene in the whole body is known to result in embryonic lethality (He et al., 2023). The conditional knockout mice were generated by Cre-loxP system. The following mouse strains were used: *Zip10*^{lox/lox} (*Zip10*^{f/f}) mouse <B6;129-Slc39a10 <tm1.1Tfk>> (RBRC06221) (Hojo et al., 2014; Miyai et al., 2014; Bin et al., 2017) was provided by the RIKEN BRC through the National BioResource Project of the MEXT/AMED, Japan. *Zip6*^{lox/lox} (*Zip6*^{f/f}) mouse <B6;129-Slc39a6> was generated by AdAMS. *Gdf9*^{Cre/+} mice <Stock Tg (GDF9-iCre) 5092Coo/J, Stock No: 011062> were delivered from Jackson Laboratory (Lan Z, et al, 2004). Crlj: C57BL/6J female mice (6-8 weeks old) and Crlj: C57BL/6J male mice (8-12 weeks old) were purchased from Charles River Laboratories Japan (Yokohama, Japan). The mice were housed under controlled lighting conditions (daily light period, 0600-1800 h). The study was approved by the Animal Experimentation Committees of Azabu University and were performed in accordance with the committees' guidelines (200318-13 and 230309-1). Oocyte-specific gene knockout (*Zip10*^{f/f}, *Gdf9*^{Cre/+}, *Zip10*^{d/d}) mice were generated by crossing *Gdf9*^{Cre/+} male mice with *Zip10*^{f/f} female mice (Fig. S1A-E). Similarly, *Zip6*^{d/d} mice were generated (Fig. S1A-E).

In situ hybridization (ISH)

ISH was performed as previously described with some modifications (Namiki et al., 2023 [DOI](#)). Briefly, fixed ovaries were paraffin-embedded and paraffin sections (6 μ m) were mounted on MAS-coated slides (Matsunami Glass Industries, Osaka, Japan) under RNase-free conditions. Sense or antisense digoxigenin (DIG)-labeled RNA probes for *Slc39a10* (*Zip10*) were purchased from Genostaff. The sections were deparaffinized, rehydrated, and post-fixed in 10% neutral buffered formalin (NBF) for 30 min at 37°C, followed by the treatment with 0.2% hydrogen chloride and 5 μ g/ml proteinase K (FUJIFILM Wako Pure Chemical, Osaka, Japan) for 10 min at 37°C, respectively. Hybridization was performed with DIG-labeled probes (250 ng/ml) in a humidified chamber at 60°C overnight. The slides were washed after hybridization, then treated with blocking reagent (Genostaff) for 15 min and alkaline phosphatase-conjugated anti-DIG antibody (1:2,000; Roche Diagnostics, Basel, Switzerland) for 1 h at room temperature. The signals were detected by 4-nitro-blue tetrazolium/5-bromo-4-chloro-3-indolyl phosphate (NBT/BCIP, Roche Diagnostics) in a humidified container for 12 h at 4°C. The sections were counterstained with Kernechtrot solution (Muto Pure Chemicals, Tokyo, Japan). Signals detected by the sense probe were used as a control for background levels.

Immunofluorescent staining (IF) for ovary

IF was performed as previously described with some modifications (Namiki et al., 2023 [DOI](#)). Fixed ovaries were paraffin-embedded and the sections (6 μ m) were deparaffinized, hydrated and conducted for antigen retrieval by autoclaving in 10 mM sodium citrate buffer (pH = 6.0) for 5 min. The sections were further incubated on ice for 30 min. After blocking with Blocking One Histo (06349-64, NACALAI TESQUE Inc, Kyoto, Japan) for 1 h, the slides were incubated with primary antibody for Rabbit anti-mouse-Zip10 (1:200; (Miyai et al., 2014 [DOI](#))), Rabbit anti-SLC39A6 (1:200, HPA042377, Sigma), Rabbit anti-Foxl2 (1:300; (Cocquet et al., 2002 [DOI](#); Polanco et al., 2010 [DOI](#))), Rat anti-ZP2 (1:100, sc-32752, Santa Cruz Biotechnology, Dallas, Texas, USA) in Can Get Signal immunostain (TOYOBO, Tokyo, Japan) overnight at 4°C. The slides were incubated with Alexa Fluor 488 donkey anti-rabbit IgG (H+L), Alexa Fluor 594 donkey anti-rabbit IgG (H+L) or Alexa Fluor 488 donkey anti-rat IgG (H+L) conjugated secondary antibodies (Jackson Immuno Research Laboratories, West Grove, PA, USA) diluted 1:500 in Can Get Signal immunostain (TOYOBO) for 1 h, and mounted with ProLong Glass Antifade Mountant with NucBlue Stain (P36981, Thermo Fisher Scientific, Waltham, MA, USA). Micrographs were captured by BZ-X700 microscopy (Keyence, Osaka, Japan).

Oocyte preparation

Germinal vesicle (GV) oocytes were collected in the manner described in our previous study (Ito et al., 2008 [DOI](#)) with some modifications. Ovaries were collected from female mice 48 h after intraperitoneally injection with 5 IU equine chorionic gonadotropin (eCG) (PMS; Nippon Zenyaku Kogyo, Fukushima, Japan). They were placed in a 35 mm culture dish containing MEM α (no nucleosides, powder; Gibco/Thermo Fisher Scientific, Tokyo, Japan) medium with 26 mM NaHCO₃, 75 mg/l penicillin, 50 mg/l streptomycin sulfate, 5% (v/v) heat-treated fetal calf serum (FCS), and 10 ng/ml epidermal growth factor (EGF). Cumulus-oocyte complexes (COCs) were released from the antral follicles by gentle puncturing with a needle.

To obtain metaphase II (MII) oocytes, the mice were intraperitoneally injected with 5 IU eCG followed by injection with 5 IU human chorionic gonadotropin (hCG) (Gonotropin; ASKA Pharmaceutical, Tokyo, Japan) at 48 h later. COCs were collected from the oviductal ampulla 14-16 h after hCG injection.

These oocytes were used in the following experiments.

Count of ovulated oocytes

COCs-MII were obtained by superovulation treatment. The cumulus cells were removed from the COCs-MII with hyaluronidase (1 mg/ml) and gentle pipetting. All collected oocytes were counted and calculated as the number of ovulations.

In vitro maturation (IVM) of GV oocytes

IVM were conducted in the manner described in our previous study (Kamoshita et al., 2021 [DOI](#)) with some modifications. Ovaries were placed in a 35 mm culture dish containing MEMα medium with 26 mM NaHCO₃, 75 mg/l penicillin, 50 mg/l streptomycin sulfate, 5% (v/v) FCS, and 10 ng/ml EGF. COCs-GV were released from the antral follicles by gentle puncturing with a needle. The COCs-GV were washed 3 times and cultured in 500 μl of same medium in 4-well dish at 37°C in an atmosphere of 5% CO₂ in air for 10, 12 and 14 h. After the culture, cumulus cells were removed from the COCs with hyaluronidase (1 mg/ml) and gentle pipetting. The extrusion of 1st polar body was evaluated at each time.

IF for oocytes/preimplantation embryos

The α-tubulin and JUNO (PE anti-mouse FR4) were performed with some modification of the methods of Inoue *et al* (Inoue et al., 2017 [DOI](#)). The oocytes or embryos were fixed in 4% PFA for 30 min at room temperature, washed in PBS containing 1% polyvinyl alcohol (PBS/PVA). They were permeabilized by treatment of 0.5% Triton X-100 for 15 min and washed 2 times in 1% BSA/PBS/PVA followed by blocked for 20 min in same medium. They were incubated for overnight at 4° with primary antibodies to rabbit anti-α-tubulin (1:200; 11H10, cell Signaling Technology) or rat anti-mouse FR4 (JUNO; 1:250; 12A5 BioLegend), wash 3 times in 1% BSA/PBS/PVA. Primary antibodies were detected using Alexa Fluor 488 donkey anti-rabbit IgG (H+L) (1:250) or Alexa Fluor 488 donkey anti-rat IgG (H+L) (1:250) for 1 h at room temperature. After staining, all samples mounted in VECTASHIELD Mounting Medium with DAPI (H-1200; Vector Laboratories, California, USA), imaged using BZ-X700 microscopy (Keyence, Osaka, Japan).

The ovastacin staining was performed with some modifications of the methods of Burkart *et al* (Burkart et al., 2012 [DOI](#)). oocytes or embryos were fixed in 4% PFA for overnight at 4 °C, washed in PBS containing 0.3% polyvinylpyrrolidone (PVP), and then blocked in 0.3% BSA/0.1 M glycine (three times for 10 min) followed by permeabilization in 0.2% Triton X-100 for 15 min (Baibakov et al., 2007 [DOI](#)). Samples were then incubated for overnight at 4°C with Rabbit polyclonal anti-ovastacin (Burkart et al., 2012 [DOI](#)), 1:200, gifted antibody), washed with 0.3% PVP/0.1% Tween (three times for 10 min), and incubated for 1 h at room temperature with Alexa Fluor 488 anti-rabbit secondary antibody (1:500) followed by staining and mounting with VECTASHIELD Mounting Medium with DAPI. Samples were imaged using TCS SP5 II confocal microscope (Leica Microsystems, Wetzlar, Germany).

In vitro fertilization (IVF)

IVF and sperm collection were conducted with some modification of the method described in our previous study (Kageyama et al., 2023 [DOI](#)). In brief, ovulated COCs-MII were preincubated for 1 h in 80 μl human tubal fluid (HTF) droplets supplemented with 1.25 mM reduced glutathione (GSH). Frozen-thawed sperm suspensions were suspended in 200 μl preincubation medium (HTF containing 0.4 mM methyl-β-cyclodextrin) and 0.1 mg/ml PVA, but without bovine serum albumin, and were incubated at 37 °C under 5% CO₂ in humidified air for 1 h. At the time of insemination, preincubated spermatozoa were transferred into the droplets with oocytes at final concentrations of 2.0×10⁶ sperm/ml. After 6 h, oocytes were separated from spermatozoa and cumulus cells using a fine glass pipette and transferred into 50 μl KSOMaa medium. They were cultured at 37 °C under 5% CO₂ in humidified air for approximately 24-96 h. The embryos were observed at 24, 48, 72, and 96h after IVF, and the number of 2cell, 4-8cell, morula, and blastocyst stage embryos were

counted, respectively. The oocytes used for IVF were calculated as the parameter for the fertilization rate and the rate of embryo development. After IVF at 96 h, the blastocysts were fixed in 4% PFA for 30 min at room temperature. Following three times washed in PBS/PVA for 30 min each. Nuclear DNA was stained and mounted in VECTASHIELD Mounting Medium with DAPI. Cell numbers were determined by visually inspecting nuclei stained with DAPI using an inverted fluorescence microscope.

Zinc measurements

FluoZin-3AM staining was conducted using the method described in our previous study (Kageyama et al., 2022 [DOI](#)). The obtained GV oocytes, cumulus cells were removed from the COCs with gentle pipetting. The obtained MII oocytes, cumulus cells were removed from the COCs with hyaluronidase (1 mg/ml) and gentle pipetting. Oocytes with a polar body were defined as MII. After IVF for 6h, two pronucleus stage embryos (2PN) were collected. GV, MII and 2PN were loaded in 50 μ l medium that was suitable for each stage containing the amyl ester of the membrane permeant zinc-specific fluorophore, FluoZin-3AM (2 μ M; F24195, Thermo Fisher Scientific, excitation 494 nm/emission 516 nm) for 1 h in humidified CO₂ [5% (v/v) in air] at 37 °C followed by washing three times in medium and then observation with TCS SP5 II confocal microscope. FluoZin-3 has been extensively characterized for measurements of free intracellular zinc in live cells using microscopy and has an affinity constant (Kd) for zinc of 15 nM (Gee et al., 2002a [DOI](#); Gee et al., 2002b [DOI](#)). Our previous study showed MII oocytes were treated with FluoZin-3 AM for 60 min, the change of fluorescence was confirmed in the cytoplasm of the oocytes and embryos, suggesting this treatment duration with FluoZin-3 AM is suitable for detection of zinc ions in oocytes and embryos (Kageyama et al., 2022 [DOI](#)). The pixel intensity per unit area after background subtraction was determined in GV, MII and 2PN within the circle (white circle) and ImageJ image processing software.

Measurement of Ca²⁺ and Zinc spark during fertilization

Fresh or frozen-thawed sperm suspension were suspended in 200 μ l preincubation medium. The 1 μ l (fresh) or 20 μ l (frozen-thawed) sperm suspension were placed in 40-80 μ l HTF drop in pre-insemination dish about 20 min before insemination. The zona pellucida were punctured by piezoelectric pulses applied to four locations with a 15 μ m injection needle. The treated MII oocytes were cultured in CZB containing CalbryteTM 590 AM (10 μ M, #20700, AAT Bioquest, CA, USA, excitation 581 nm/emission 593 nm) for 10 min in humidified CO₂ [5% (v/v) in air] at 37 °C. While the oocytes are incubated for 10 min, make a drop of 10 μ l of PVA(-), Ca(-) Hepes-CZB (H-CZB) and pull out 7 μ l of medium from drop with pipette man in the insemination dish. After 10 min, the oocytes are washed and transferred to insemination dish and the oocytes are attached to the dish. After attachment, an HTF containing 7 μ l of BSA and a membrane-impermeable zinc-specific fluorophore, FluoZin-3 (20 μ M, F24194, Thermo Fisher Scientific, excitation 494 nm/emission 516 nm), is added to the 3 μ l drop containing the oocytes gently using a capillary. The about 1 μ l sperm were sucked from the pre-insemination drop with the capillary under a stereomicroscope and place into the drop containing the oocytes, to start imaging. In case ionomycin treatment, after calcium labeling, MII were allowed to settle in 45 μ l PVA(-), Ca(-) H-CZB containing FluoZin-3 medium drop on dish, to start imaging. The 5 μ l ionomycin (5 μ M; #407950) were added to MII at 2 min after imaging start. Imaging was performed on confocal microscope using 488 nm (Zn²⁺) and 555 nm (Ca²⁺) excitation (Nikon solutions, Tokyo, Japan) for max 50 min every 4 seconds. Imaging analysis was performed by defining regions of interest (ROIs) and measuring fluorescence intensity over time using NIS-Elements (Nikon). The intracellular ROIs were drawn as the entire interior area of the cell. The extracellular ROIs were defined as a ring around the perimeter of cell. The ring thickness was conserved for all data analyses.

Western blotting

Western blotting was carried out as described (Ito et al., 2010) with some modifications. Thirty MII oocytes or 2PN embryos were lysed in Laemmli sample buffer (Bio-Rad Laboratories, Tokyo, Japan) with 5% 2-mercaptoethanol. Samples were separated on 8% Bis-Tris gels by SDS-PAGE and transferred to PVDF membranes (Bio-Rad). The PVDF membranes were blocked in 10% skim milk (FUJIFILM Wako) in Tris-buffered saline (TBS) with 0.1% Tween-20 (Yoneyama yakuhin kogyo, Osaka, Japan) and probed with primary antibody to Rabbit anti-mouse-Zip10 (1:1,000; (Miyai et al., 2014)), Rabbit anti-SLC39A6 (1:1,000, HPA042377), rat anti-ZP2 (1:1,000) or monoclonal mouse anti- β -actin (1:5,000; A5316, Sigma-Aldrich) for overnight at 4°C. The membranes were incubated with secondary antibody; HRP-conjugated anti-rabbit IgG (1:5,000; Cell Signaling Technology, Danvers, MA, USA), HRP-conjugated anti-rat IgG (1:5000; Cell Signaling Technology) or HRP-conjugated anti-mouse IgG (1:5,000; Cell Signaling Technology) for 1 h at room temperature. After washing of the membranes, immunoreactive proteins were visualized using ECLTM Western Blotting Analysis System (Cytiva Global Life Sciences Technologies, Tokyo, Japan), according to the manufacture's recommendation. The membranes were exposed by ImageQuant LAS 4000. After exposure, the membranes were incubated for 30 min at 50 °C in the stripping buffer including 1.5 M Tris-HCl (pH 6.8), 10% SDS, 2-mercaptoethanol followed by extensive washing of the membranes. The membranes were probed with primary antibody to mouse anti- β -actin (1:5000; Sigma-Aldrich) for overnight at 4°C. The membranes were incubated with secondary antibody; HRP-conjugated anti-mouse IgG (1:5000; Cell Signaling) for 1 h at room temperature. The same procedure was followed below. The intensities of ZP2 bands were measured by quantitative analysis by densitometry using Image J.

Statistical analysis

Values from three or more times were used for evaluation of statistical significance. Statistical analysis was performed using Statcel 3 software (OMS Ltd., Saitama, Japan). The fertility, the total numbers of oocytes collected from each mouse, the percentage of first polar extrusions, the fluorescence intensity of FluoZin-3 AM, the cell count of blastocysts and Quantification of JUNO expression were evaluated statistically by Student's or Welch's *t*-test analysis. The rate of fertilization and embryo development was analyzed using chi square tests. The Quantification of ZP2 expression before and after fertilization was analyzed using one-way ANOVA. Values are shown as means $\pm$ S.E.M, and significant differences were considered at *p* values < 0.05.

Data and materials availability

All data are available in the main text or the supplementary materials.

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Additional information

Author contributions

Conceptualization: A.K., J.I., Data curation: A.K., Methodology: N.O., T.W., A.O., Investigation: A.K., N.O., T.N., Y.K., Resources: M.O., Y.Y., T.F., Visualization: A.K., N.O., T.N., Supervision: A.O., T.F., R.A.F. J.I., N.K., Writing—original draft: A.K., Writing—review & editing: All the authors reviewed and edited the manuscript.

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Additional files

Supplementary information

Movie 1. Monitoring of intracellular calcium ions during fertilization of Zip10^{ff} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip10^{ff} oocytes displayed the calcium oscillations following fertilization. The changes in zinc ions were also monitored simultaneously (Movie 2). The video was excerpted from maximum 50 min. 

Movie 2. Monitoring of extracellular zinc ions during fertilization of Zip10^{ff} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip10^{ff} oocytes released zinc ions into the extracellular environment through a zinc spark followed the first calcium rise following fertilization. The changes in

calcium ions were also monitored simultaneously (Movie 1). The video was excerpted from maximum 50 min. [↗](#)

Movie 3. Monitoring of intracellular calcium ions during fertilization of Zip10^{d/d} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip10^{d/d} oocytes displayed the calcium oscillations following fertilization. The changes in zinc ions were also monitored simultaneously (Movie 4). The video was excerpted from maximum 50 min. [↗](#)

Movie 4. Monitoring of extracellular zinc ions during fertilization of Zip10^{d/d} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip10^{d/d} oocytes did not release zinc ions into the extracellular environment followed the first calcium rise following fertilization. The changes in calcium ions were also monitored simultaneously (Movie 3). The video was excerpted from maximum 50 min. [↗](#)

Movie 5. Monitoring of intracellular calcium ions during parthenogenesis of Zip10^{f/f} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip10^{f/f} oocytes displayed a transient rise of calcium ions following artificial oocyte activation with ionomycin. The changes in zinc ions were also monitored simultaneously (Movie 6). The video was excerpted from maximum 50 min. [↗](#)

Movie 6. Monitoring of extracellular zinc ions during parthenogenesis of Zip10^{f/f} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip10^{f/f} oocytes released zinc ions into the extracellular environment through a zinc spark followed a transient calcium rise following artificial oocyte activation with ionomycin. The changes in calcium ions were also monitored simultaneously (Movie 5). The video was excerpted from maximum 50 min. [↗](#)

Movie 7. Monitoring of intracellular calcium ions during parthenogenesis of Zip10^{d/d} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip10^{d/d} oocytes displayed a transient rise of calcium ions following artificial oocyte activation with ionomycin. The changes in zinc ions were also monitored simultaneously (Movie 8). The video was excerpted from maximum 50 min. [↗](#)

Movie 8. Monitoring of extracellular zinc ions during parthenogenesis of Zip10^{d/d} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip10^{d/d} oocytes did not release zinc ions into the extracellular environment followed a transient calcium rise following artificial oocyte activation with ionomycin. The changes in calcium ions were also monitored simultaneously (Movie 5). The video was excerpted from maximum 50 min. [↗](#)

Movie S1. Monitoring of intracellular calcium ions during fertilization of Zip6^{f/f} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip6^{f/f} oocytes displayed the calcium oscillations following fertilization. The changes in zinc ions were also monitored simultaneously (Movie S2). The video was excerpted from maximum 50 min. [↗](#)

Movie S2. Monitoring of intracellular calcium ions during fertilization of Zip6^{f/f} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip6^{f/f} oocytes released zinc ions into the extracellular environment through a zinc spark followed the first calcium rise following fertilization. The changes in calcium ions were also monitored simultaneously (Movie S1). The video was excerpted from maximum 50 min. [↗](#)

Movie S3. Monitoring of intracellular calcium ions during fertilization of Zip6^{d/d} oocytes. The changes in Ca²⁺ were detected by CalbryteTM 590 AM every 4 second. Zip6^{d/d} oocytes displayed the calcium oscillations following fertilization. The changes in zinc ions were also monitored simultaneously (Movie S4). The video was excerpted from maximum 50 min. [↗](#)

Movie S4. Monitoring of intracellular calcium ions during fertilization of Zip6^{d/d} oocytes. The changes in Zn²⁺ were detected by FluoZin-3 every 4 second. Zip6^{f/f} oocytes released zinc ions into the extracellular environment through a zinc spark followed the first calcium rise following fertilization. The changes in

calcium ions were also monitored simultaneously (Movie S3). The video was excerpted from maximum 50 min. [🔗](#)

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Reviewer #1 (Public review):

The revised manuscript addresses several reviewer concerns, and the study continues to provide useful insights into how ZIP10 regulates zinc homeostasis and zinc sparks during fertilization in mice. The authors have improved the clarity of the figures, shifted emphasis in the abstract more clearly to ZIP10, and added brief discussion of ZIP6/ZIP10 interactions and ZIP10's role in zinc spark-calcium oscillation decoupling. However, some critical issues remain only partially addressed.

(1) Oocyte health confound: The use of Gdf9-Cre deletes ZIP10 during oocyte growth, meaning observed defects could result from earlier disruptions in zinc signaling rather than solely from the absence of zinc sparks at fertilization. The authors acknowledge this and propose transcriptome profiling as a future direction. However, since mRNA levels often do not accurately reflect protein levels and activity in oocytes, transcriptomics may not be particularly informative in this context. Proteomic approaches that directly assess the molecular effects of ZIP10 loss seem more promising. Although current sensitivity limitations make proteomics from small oocyte samples challenging, ongoing improvements in this area may soon allow for more detailed mechanistic insights.

(2) ZIP6 context and focus: The authors clarified the abstract to emphasize ZIP10, enhancing narrative clarity. This revision is appropriate and appreciated.

(3) Follicular development effects: The biological consequences of ZIP6 and ZIP10 knockout during folliculogenesis are still unknown. The authors now say these effects will be studied in the future, but this still leaves a major mechanistic gap unaddressed in the current version.

(4) Zinc spark imaging and probe limitations: The addition of calcium imaging enhances the clarity of Figure 3. However, zinc fluorescence remains inadequate, and the authors depend solely on FluoZin-3AM, a dye known for artifacts and limited ability to detect subcellular labile zinc. The suggestion that C57BL/6J mice may differ from CD1 in vesicle appearance is plausible but does not fully address concerns about probe specificity and resolution. As the authors acknowledge, future studies with more selective probes would increase confidence in both the spatial and quantitative analysis of zinc dynamics.

(5) Mechanistic insight remains limited: The revised discussion now recognizes the lack of detailed mechanistic understanding but does not significantly expand on potential signaling pathways or downstream targets of ZIP10. The descriptive data are useful, but the inability to pinpoint how ZIP10 mediates zinc spark regulation remains a key limitation. Again, proteomic profiling would probably be more informative than transcriptomic analysis for identifying ZIP10-dependent pathways once technical barriers to low-input proteomics are overcome.

Overall, the authors have reasonably revised and clarified key points raised by reviewers, and the manuscript now reads more clearly. However, the main limitation, lack of mechanistic insight and the inability to distinguish between developmental and fertilization-stage roles of ZIP10, remains unresolved. These should be explicitly acknowledged when framing the conclusions.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

The authors investigated the role of the zinc transporter ZIP10 in regulating zinc sparks during fertilization in mice. By utilizing oocyte-specific Zip6 and Zip10 conditional knockout mice, the authors effectively demonstrate the importance of ZIP10 in zinc homeostasis, zinc spark generation, and early embryonic development. The study is overall useful as it identifies ZIP10 as an important component of oocyte processes that support embryo development, thus opening the door for further investigations. While the study provides solid evidence for the requirement of ZIP10 in the regulation of zinc sparks and zinc homeostasis, it falls short of revealing the underlying mechanism of how ZIP10 exerts this important function.

This report is the first to clarify the role of the zinc transporters ZIP10 expressed in oocytes, which was previously unknown, and does not focus on the detailed mechanism. As you pointed out, we believe that the mechanism will also be important information in the field of fertilization and embryogenesis research, and we believe that it is necessary to consider this issue in the future.

(1) The zinc transporters the authors are knocking out are expressed in mouse oocytes through follicular development, and the Gdf9-cre driver used means these oocytes were grown in the absence of appropriate Zinc signaling. Thus, it would be difficult to assert that the lack of fertilization associated with zinc sparks is solely responsible for the failure of embryo development. Spindle morphology and other meiotic parameters do not necessarily report oocyte health, so normalcy of these features may not be a strong argument when it comes to metabolic issues.

As you rightly observe, the results of this study do not entirely exclude the possibility of oocyte health in the absence of adequate zinc homeostasis during oocyte growth. However, evidence has been presented demonstrating that spindle formation does occur in Zip10^{d/d} mouse oocytes (Fig.2 C), that fertilization occurs despite the absence of zinc spark (Fig.3 and Fig. 4A), and that some embryos develop to blastocysts (Fig. 4 B). We believe that future studies should evaluate the transcriptome profile of Zip10^{d/d} mouse oocytes.

(2) While comparing ZIP6 and ZIP10 in the abstract provides context, focusing more on ZIP10 would improve reader comprehension, as ZIP10 is the primary focus of the study. Emphasizing the specific role of ZIP10 will help the reader grasp the core findings more clearly.

Thank you for your valuable input. We have revised the summary to focus more on ZIP10 by removing the section in the summary that mentions ZIP6 (P.1-2 Line 34-52).

(3) Zinc transporters ZIP6 and ZIP10 are expressed during follicular development, but the biological significance of the observation is not clearly addressed. The authors should investigate whether the ZIP6 and ZIP10 knockout affects follicular development and discuss the potential implications.

Thank you for your valuable input. As you mentioned, we have not been able to clarify the effects of ZIP6 and ZIP10 knockout on follicle formation. However, this report clarifies the role of ZIP-mediated zinc ions in their inclusion. The effect of ZIP knockout on follicle formation will be discussed in the future.

(4) In Figure 3, the zinc fluorescence images are unclear, making it difficult for readers to interpret the data. Including snapshot images of calcium and zinc spikes as part of the main figure would improve clarity. Moreover, adding more comparative statements and a deeper explanation of why Zip10 KO mice exhibit normal calcium oscillations but lack zinc sparks would strengthen the manuscript.

Thank you for your suggestion. We have also added images of calcium elevation after fertilization to Fig. 3 and Fig. S3. In addition, the figure legends have been changed (P.29 Line 937-939, P.34 Line 1104-1106). As to why Zip10 KO mice show normal calcium oscillations but lack zinc spikes, as mentioned in Discussion (P. 10 Line 299-300), we speculate that zinc ions existed in Zip10d/d mouse oocytes induce Ca^{2+} release without compromising IP3R1 sensitivity. We also assume that the lack of zinc spark is due to low accumulation of zinc ion levels in the oocytes via ZIP10, as described in Discussion (P.10 Line 300-302).

(5) While the study identifies the role of ZIP10 in zinc spark generation, it lacks a clear mechanistic insight. The topic itself is interesting, but without providing a more detailed explanation of the underlying mechanisms, the study leaves an important gap. Further discussion on the signaling pathways potentially involved in zinc spark regulation would add depth to the findings.

Thank you for your input. This report is the first to clarify the role of the zinc transporters ZIP6 and ZIP10 expressed in oocytes, which was previously unknown, and does not focus on the detailed mechanism. As you pointed out, we believe that the mechanism and signaling pathways will also be important information, and we believe that it is necessary to research this issue in the future.

Reviewer #2 (Public review):

Summary:

In this important study, the authors examine the role of two zinc uptake transporters, Zip6 and Zip10, which are important during the maturation of oocytes, and are critical for both successful fertilization and early embryogenesis.

Strengths:

The authors report that oocytes from Zip10 knockout mice exhibit lower labile zinc content during oocyte maturation, decreased amounts of zinc exocytosis during fertilization, and affect the rate of blastocyst generation in fertilized eggs relative to a control strain. They do not observe these changes in their Zip6 knockout animals. The authors present clear and well-documented results from a broad range of experimental modalities in support of their conclusions.

Thank you for your positive comments.

Weaknesses:

(1) The authors' statement that Zip10 is not expressed in the oocyte nuclei (line 252). Furthermore, in that study, ZIP10 was detected in the nuclear/nucleolar positions of oocytes of all follicular stages (Chen et al., 2023), which we did not observe. This is not supported by Figure 1, where some Zip10 signal is apparent in the primordial, primary, and secondary follicle oocytes. This statement should be corrected.

Thank you for pointing this out. Our results of ISH staining (Fig. 1A) and immunofluorescence staining (Fig. 1B) showed that it was not detected at the nucleus/nucleolus location. In other words, they could not be detected at the mRNA and protein levels. Based on the results of ISH staining and immunofluorescence staining, we conclude that it is expressed in the plasma membrane.

(2) Based on the FluoZin-3AM data, there appears to be less labile zinc in the Zip10d/d oocyte, eggs, and embryos; however, FluoZin-3AM has a number of well-known artifacts and does not accurately capture the localization of labile zinc pools. The patterns do not correspond to the well-documented zinc-containing cortical vesicles. Another zinc probe, such as ZinPyr-4 or ZincBY-1 should be used to visualize the zinc vesicles and confirm that there is less labile zinc in these locations as well.

Thank you for your suggestion. Previous studies (Lisle et al., 2013, Reproduction) and our report (Kageyama et al., 2022, Animal Science Journal) have shown that it is possible to examine the presence of labile zinc ions in oocytes and embryos. In addition, mouse oocytes (embryos) reported in previous studies are from CD1 (ICR) mice, whereas our study was conducted using C57BL/6J mice. In our report (Kageyama et al., 2024, Journal of Reproduction and Development), we reported that the appearance of zinc vesicles in the oocytes observed by FluoZin-3AM staining in CD1 and C57BL/6J mice is different, and we believe that this appearance of cortical vesicles in C57BL/6J mice is not a problem. As you say, we have not used other zinc probes and will consider this in the future.

(3) Line 268 The results indicate that ZIP10 is mostly responsible for the uptake of zinc ions in mouse oocytes. The situation seems a bit more complicated given that the differences in labile zinc content between oocytes from the WT and Zip10d/d animals are small (only 20-30 %) and that the zinc spark is diminished but still apparent at a low level in the Zip10d/d oocytes. Clearly, other factors are involved in zinc uptake at these stages. A variety of studies have suggested that Zip6 and Zip10 work together, perhaps even functioning as a heterodimer in some systems. The double KO would address this more clearly, but if it is not available, it might be more prudent to state that Zip10 plays some role in uptake of zinc in mouse oocytes while the role of Zip6 remains uncertain.

We would like to express our gratitude for the comments received. The phenotype of double knockout mice for ZIP6 and ZIP10 will be discussed at a future date. We have also added to the text that the role of ZIP6 remains uncertain (P. 11 Line 353-354).

(4) Zip6d/d oocytes did not have changes in labile zinc, nor did the lack of Zip6 have an impact on the zinc spark. However, Figure S1 does show a small amount of detectable Zip6 in the western blot. It is possible that this small amount could compensate for the complete lack of Zip6. Can ZIP6 be found in immunofluorescence of GV oocytes or MII eggs from the Zip6d/d animals? Additionally, it is possible that Zip6's role is only supplementary to that of Zip10. The authors should discuss this possibility. It would also be interesting to see if the Zip6/Zip10 double knockout displays greater defects compared to the Zip10 knockout when considering previous studies.

Thank you for your input. The mice are deficient in the gene so that ZIP6 is not functional. It is our notion that the results of WB analysis are not indicative of protein structural functionality, even in cases where the ZIP6 antibody detects a small amount of protein. Since the role of ZIP6 was not elucidated in this study, we added a statement to that effect in the text (P. 11 Line 353-354). In addition, studies using ZIP6/Zip10 double knockout mice will be discussed in the future.

Recommendations for the authors: please note that you control which revisions to undertake from the public reviews and recommendations for the authors.

We have revised the text based on the reviewer's suggestions.

Reviewer #1 (Recommendations for the authors) :

(1) In lines 133-136, it seems that the authors would like to aim to emphasize the lack of research on oocytes compared to other tissues and cells. However, the inclusion of unrelated contexts, such as the role of ZIP10 in cancer and skin, appears unnecessary and detracts from the focus on oocyte-specific mechanisms. Removing these unrelated sentences would help maintain clarity and relevance in the introduction.

*As you indicated, we removed the sentence that is not related to oocytes (P.4 Line 120-125). Further, they reported that targeted disruption using Zip6- and Zip10- specific morpholino injection or antibody incubation induced alteration of the intracellular labile zinc content, spontaneous resumption of meiosis from the PI arrest and premature arrest at a telophase I-like state (Kong et al., 2014). It is clear from these reports that ZIP6 and ZIP10 are involved in zinc transport in oocytes, but the function is not elucidated."

(2) Ensure that all video files are properly labeled to enhance understanding.

Improved video labels for clarity (Movie 1-8, Movie S1-S4)

(3) Correct mislabeling issues, such as the one in line 209.

Corrected as follows: Zip10^{d/d} mouse oocytes can be fertilized but were unlikely to develop to blastocysts (P. 6-7 Line 196-197).

(4) In Figure 4D, the amount of ZIP2 appears to increase relative to actin. Including quantification would make the data more robust. Similarly, in Figure 4F, JUNO levels appear increased in Zip10 KO. Please provide quantification.

The WB band images in Fig. 4D were quantified and their graphs were added to lower part of Fig. 4D. Furthermore, the Juno of Immunofluorescent images in Figure 4F were quantified and their graphs were added to Fig. S4. Figure legends and text were corrected and added. P. 30 Line 975-979: Expression level of β -actin serves as a protein loading control and quantified

the expression level of ZP2. Molecular mass is indicated at the left. Statistical differences were calculated according to the one-way ANOVA. Different letters represent significant differences ($p < 0.05$).

P. 35 Line: Fig. S4 Comparison of JUNO expression in Zip10^{f/f} and Zip10^{d/d} mouse MII oocytes. To measure JUNO-immunofluorescence intensity, oocytes images were selected as regions of interest (ROIs) and measured using ImageJ. Statistical differences were calculated according to student's t-test ($p > 0.05$; no significant difference). P.7 Line 206-209: As for the expression of JUNO, it had the same expression than between null and control oocytes (Fig. S4) and the temporal dynamics of its disappearance from the cortex after fertilization was similar for both Zip10^{f/f} and Zip10^{d/d} groups (Fig. 4F).

(5) Some of the sentences lack proper references.

The entire text was reviewed and references inserted where necessary.

P.7 Line 221, P.7 Line 222-223, P.8 Line 253-254, P.12 Line 358-360 and P.24 Line 698-699.

Reviewer #2 (Recommendations for the authors) :

Revisions are warranted in order to address the issues noted in the Weaknesses section of the Public Review.

Thank you for your comments, we have individually addressed the areas you pointed out in the Weaknesses section. The following text has also been corrected and edited.

(1) Line 247 "In primordial follicles, the ooplasmic staining of ZIP10 we anticipate corresponds to ooplasmic vesicular sites.

The text of P. 8 Line 230-232 was revised as follows.

"In primordial follicles, the ooplasm staining of ZIP10 we anticipate corresponds to ooplasmic vesicular sites.

(2) Line 926 "ZP2 was not stained in primordial follicle, but primary, secondary, and antral follicles stained. FOXL2 was observed in granulosa cells in 928 of all stage follicles. The scale bar represents 20 μm of primordial-secondary follicle and 150 μm of antral follicle." All three sentences have grammar issues that should be fixed.

The text of p.28 Line 908-911 was revised as follows.

It was observed that ZP2 was not present in the primordial follicle; however, it was present in the primary, secondary and antral follicles. Furthermore, FOXL2 was observed at granulosa cells of all stage follicles. Scale bar: 20 μm (primordial, primary and secondary follicle); 150 μm (antral follicle).

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